

Today $\Rightarrow$ 28 Sep^t
= $\hookrightarrow$ DP week 1
Doubts

Doubt class with Lakshay

Special class

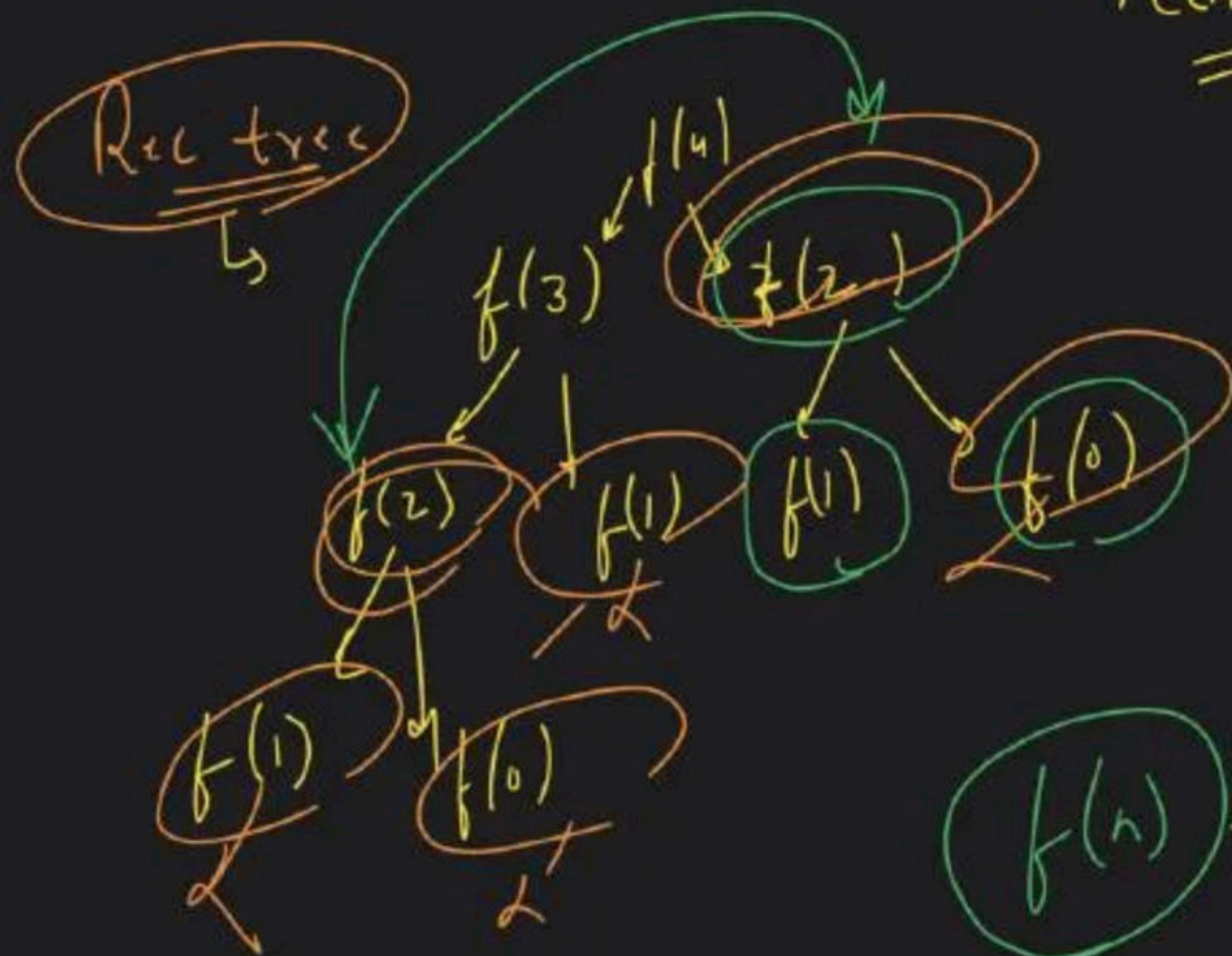
29 Sept $\rightarrow$ 9 PM
 $\hookrightarrow$ DP W2
Doubts



DP Class - 1

Special class

→ DP → Dynamic Programming
Technique



when
to apply

- DP Lagged eye
- (a) Overlapping Subproblems
 - (b) Optimal Substructure

$$f(n) = f(n-1) + f(n-2)$$

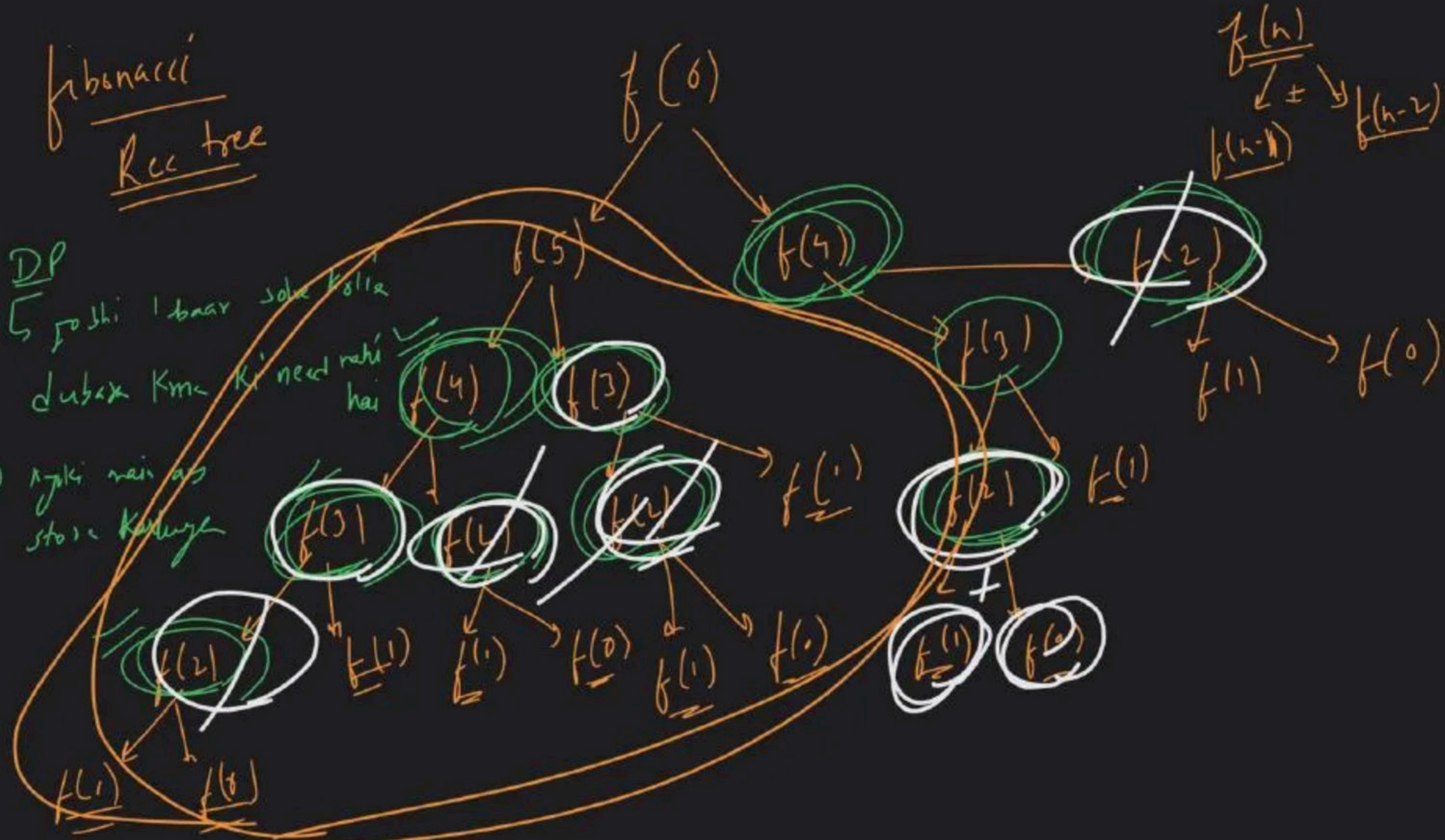
fibonacci Rec tree

DP

↳ jo shi 1 baar solve hoia

dubara kmc ki need nahi hai

↳ agle mai us store karunga



Amazon

is issue

apply

same
item

apply

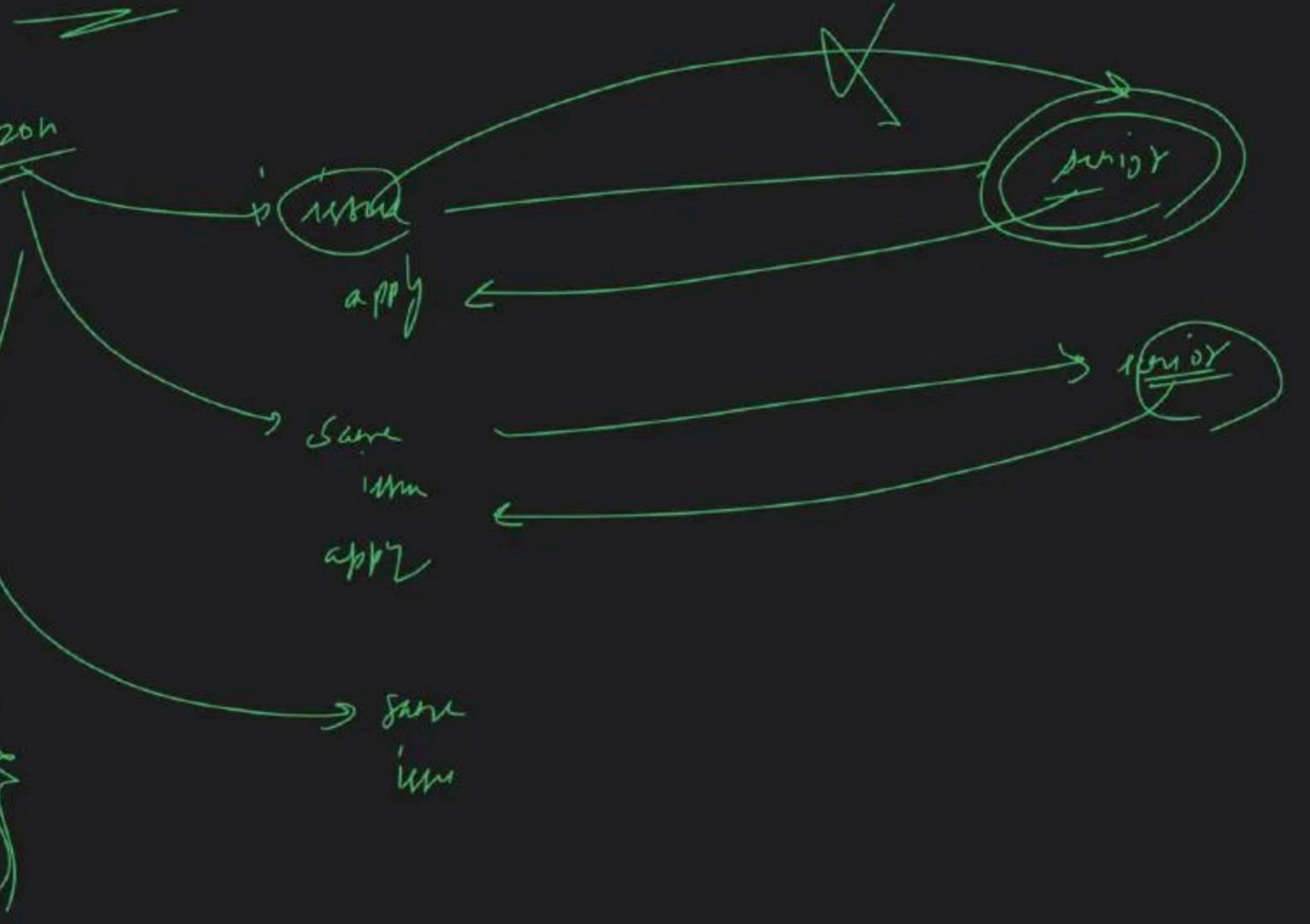
same
item

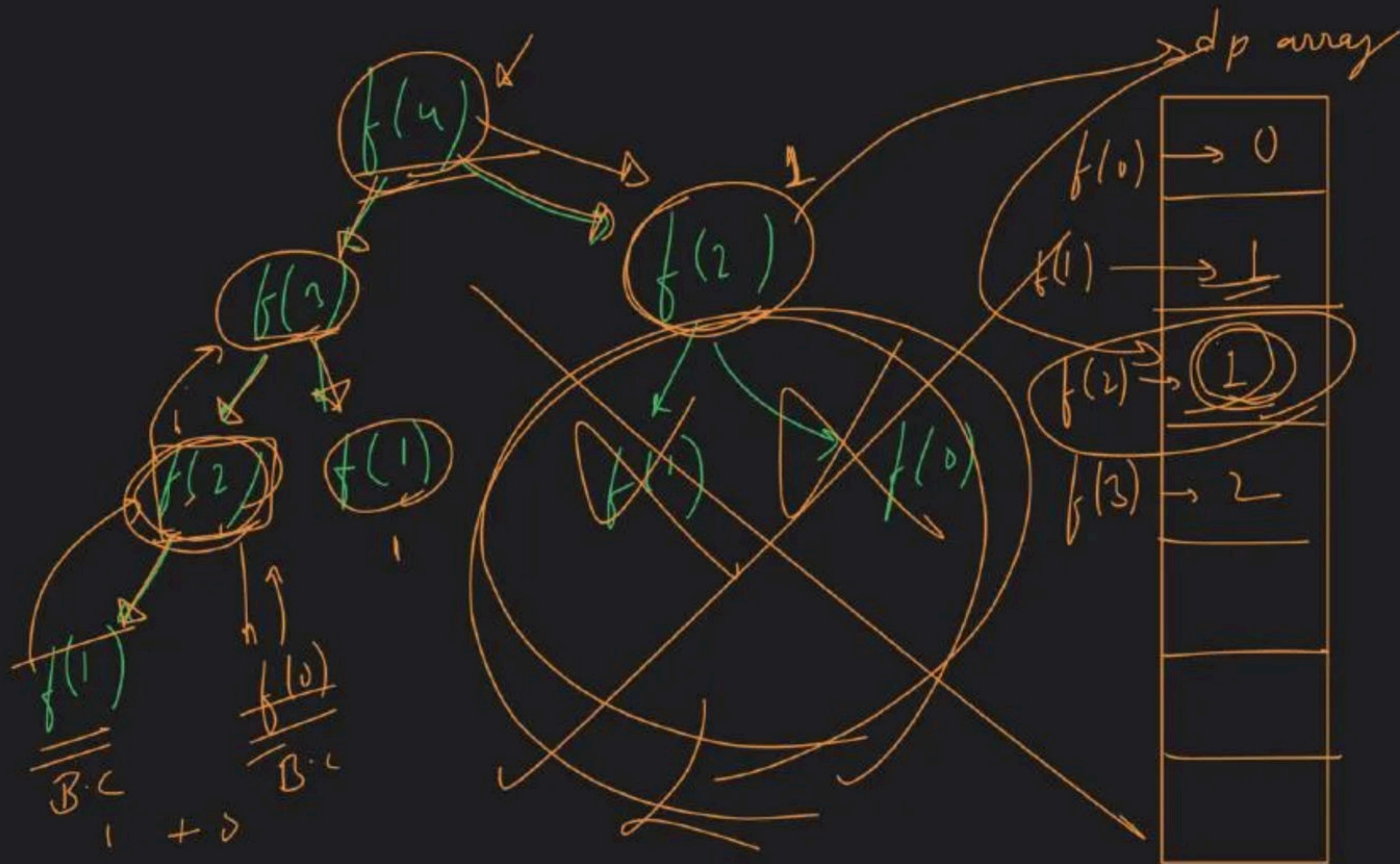
senior

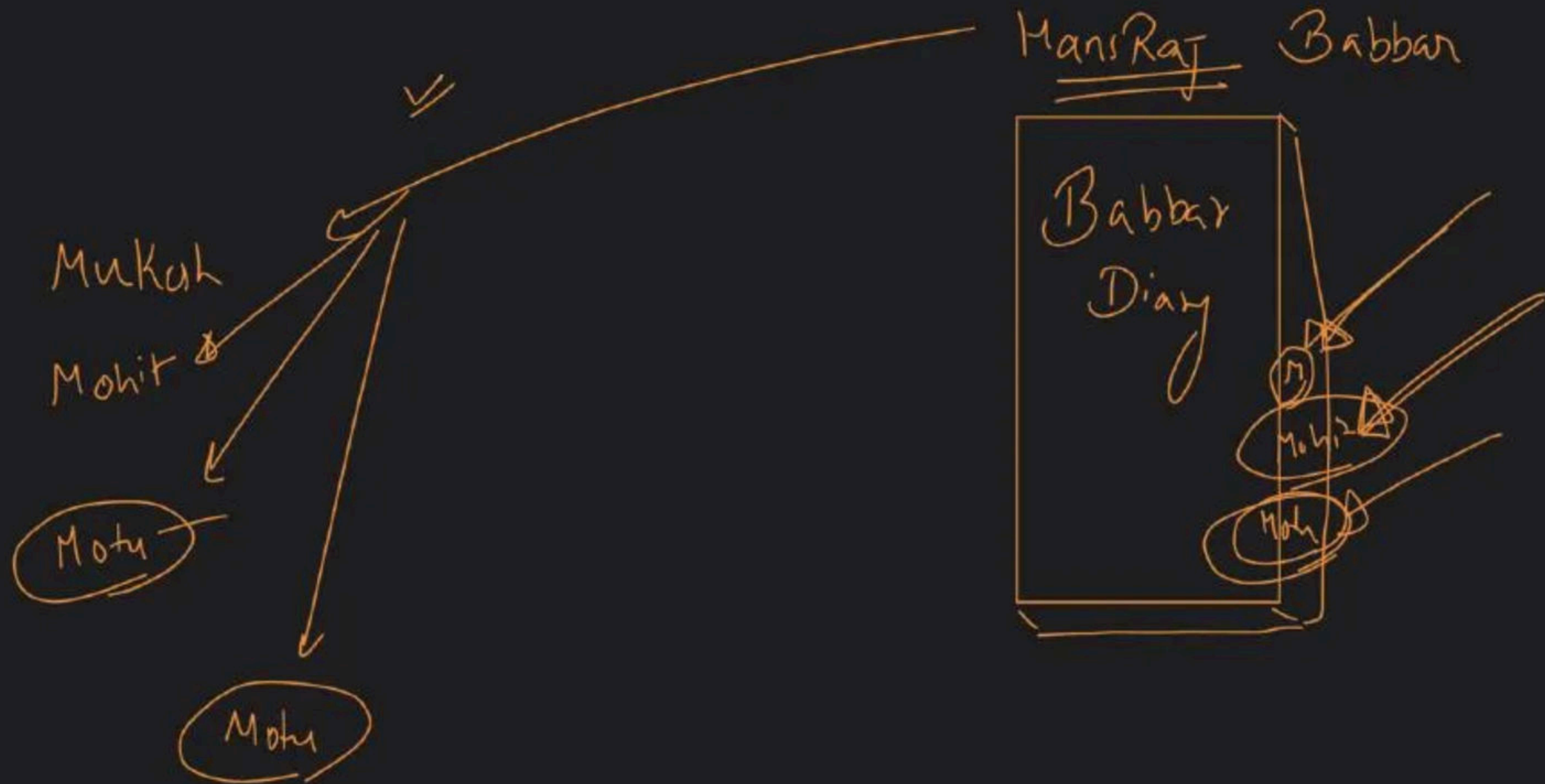
senior

sheet
with

item
copy
in
row







hanakya



galth

for Kashi
na dohraye

→ Rajya



dohraye

→ lallu

DP

Question

→ ①
→ ②
→ ③

Top-down approach

Rec +
memoisation

Bottom-up approach

loop + D.S

Space Optimisation

if possible

Pattern find

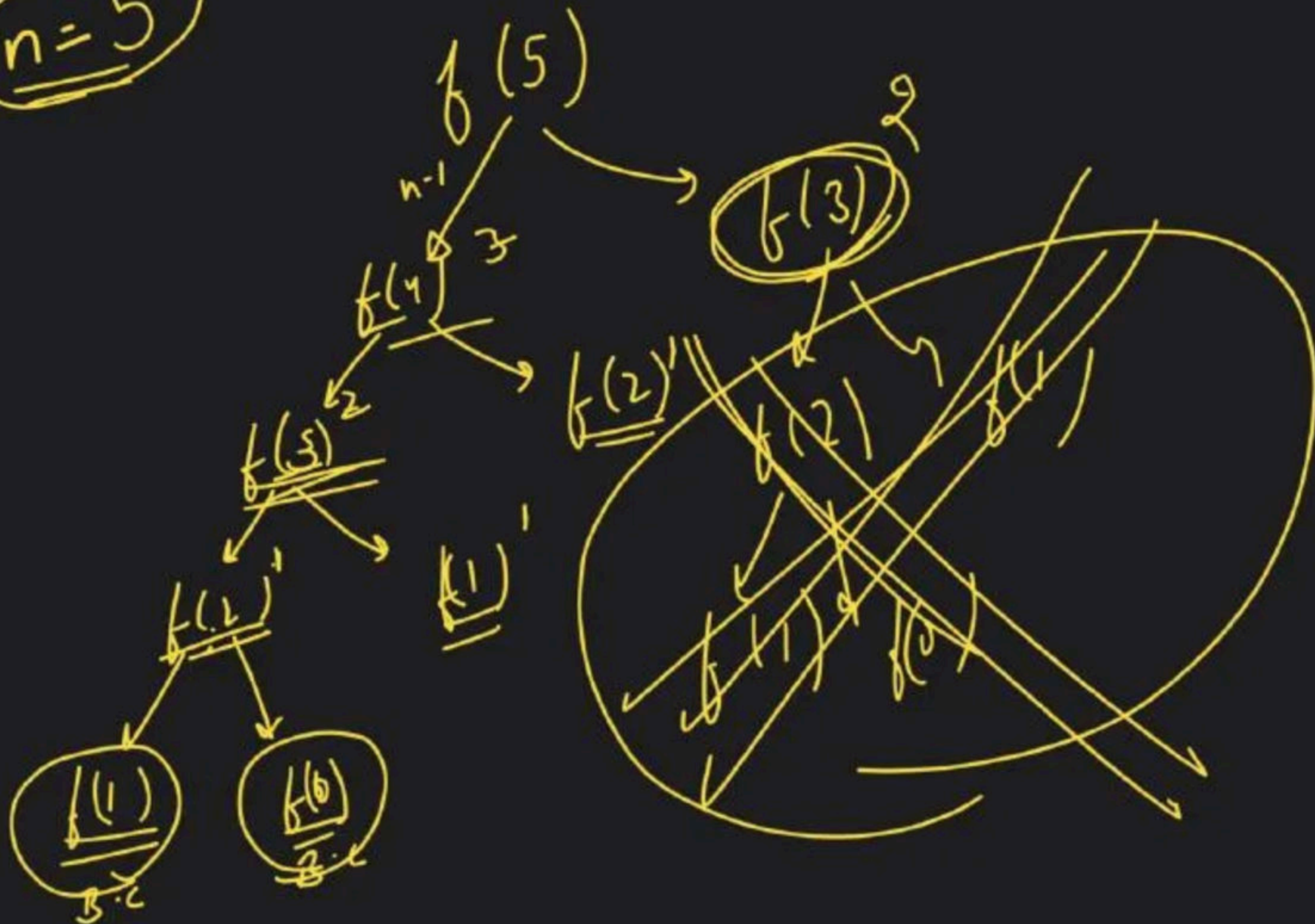
Top-down → Rec + Mem

① find out → 1D/2D/3D dp
create dp array → func pass

② ans Rec code → store in dp array & return

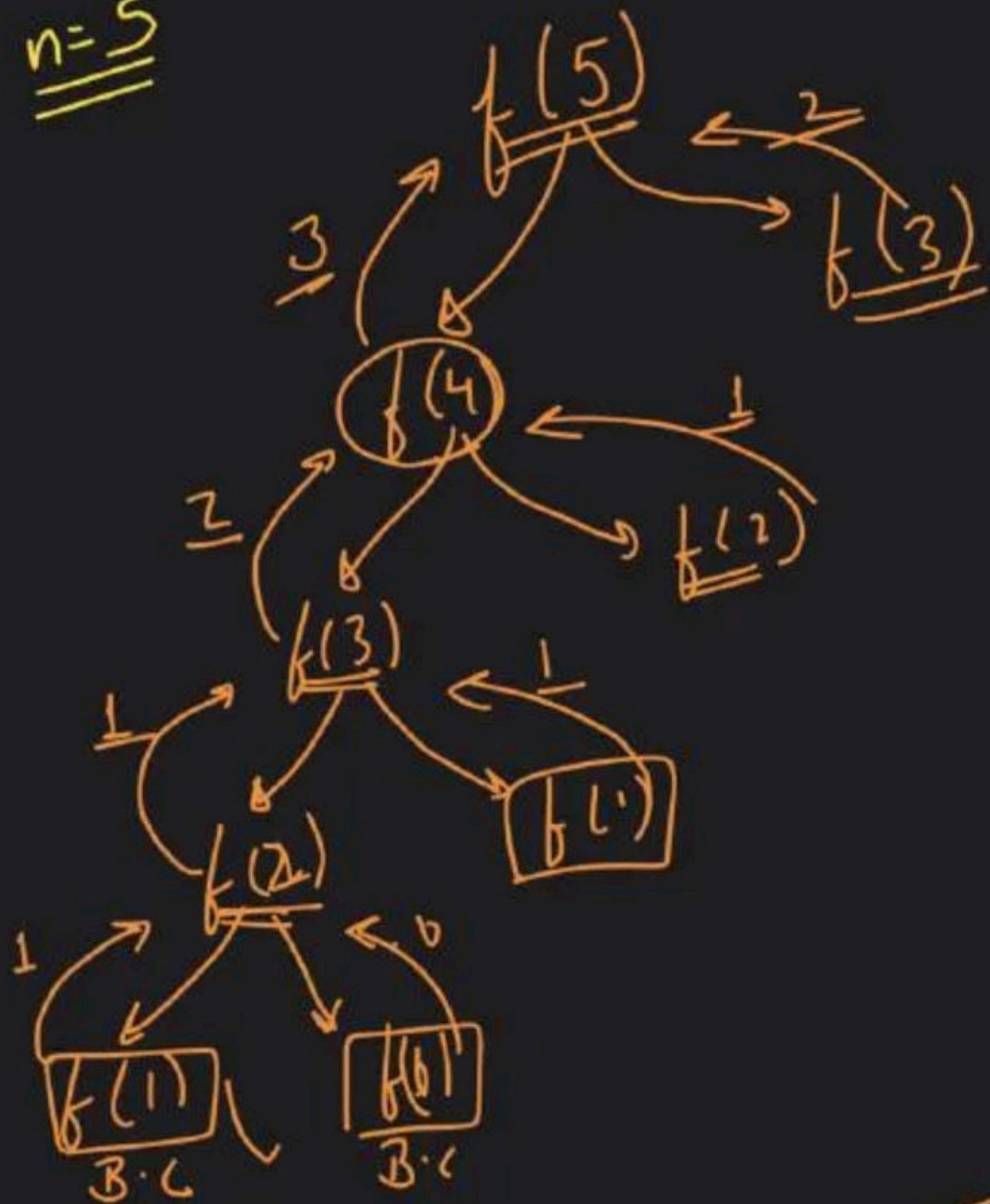
③ if ans already exist in DP array → no need to solve again → just return the ans

$n=5$



dp	
0	0
1	1
1	2
2	3
3	5
5	

n=5

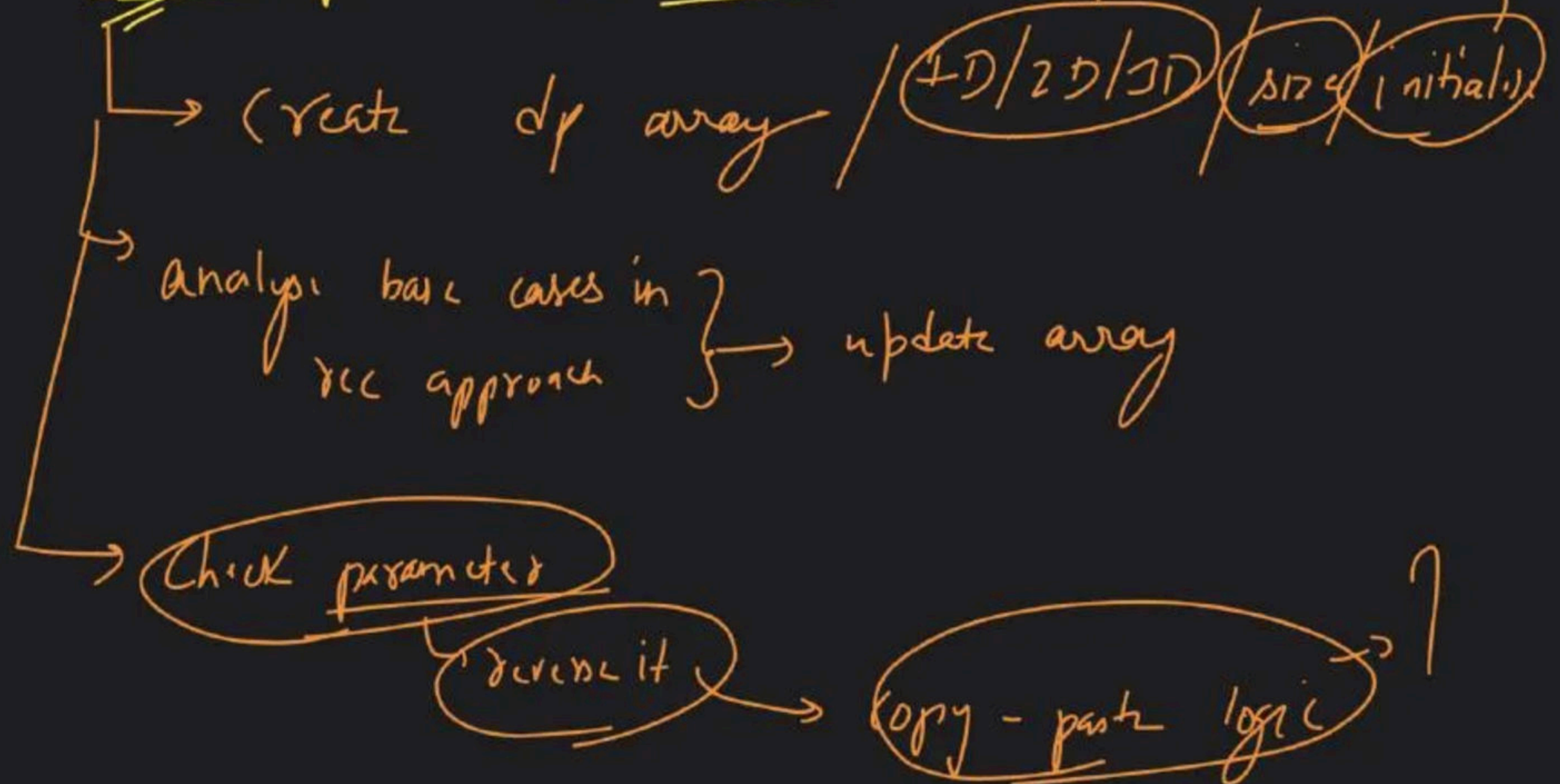


dp

1
1
1
2
3
5

0 =
1 =
2 =
3 =
4 =
5 =

→ Bottom-Up → Iterative



→ Space Optimisation :-

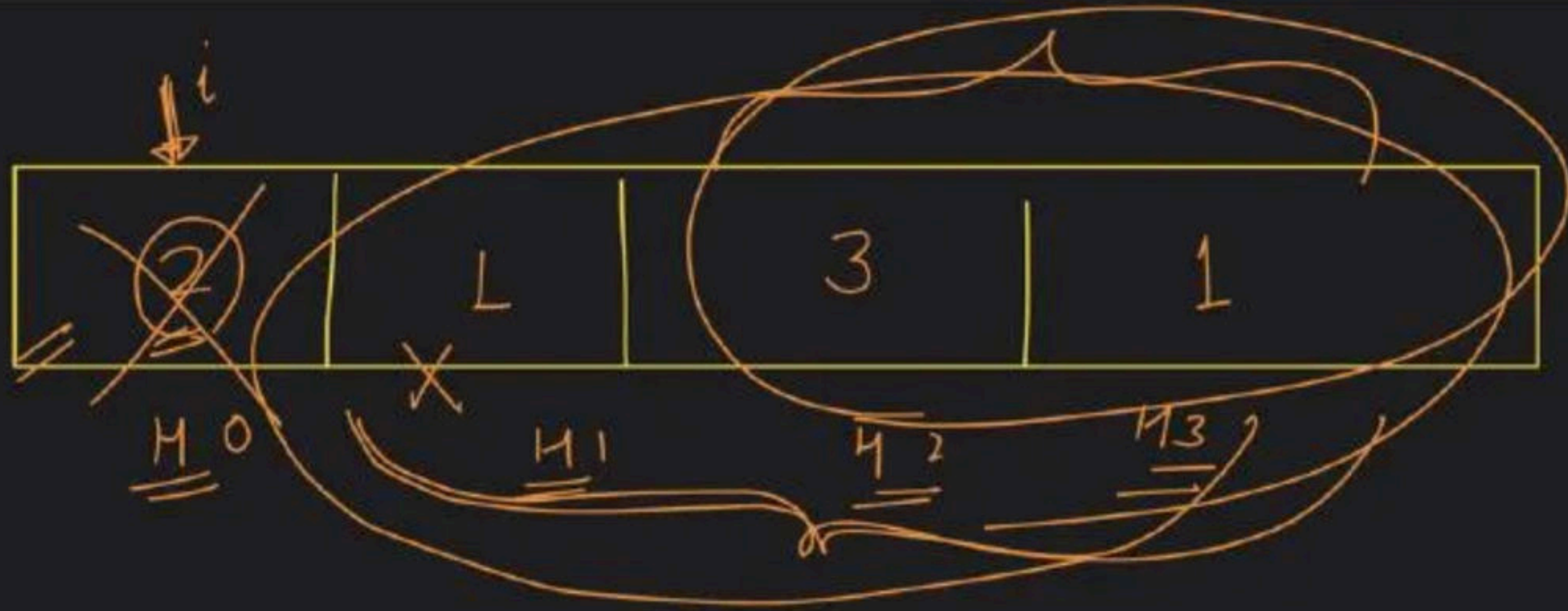
Kal

Bottom-up

Pattern

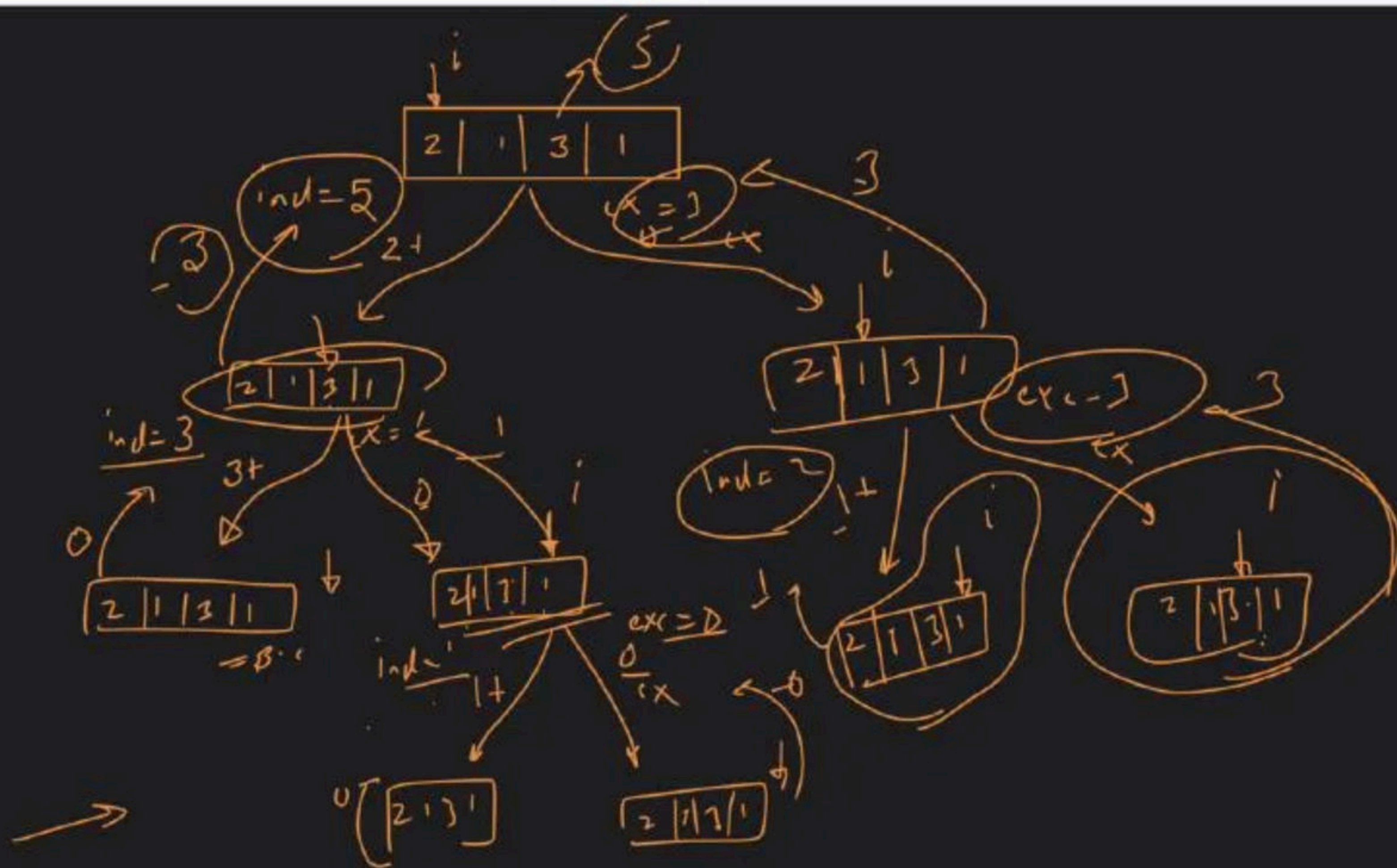
DP Class - 2

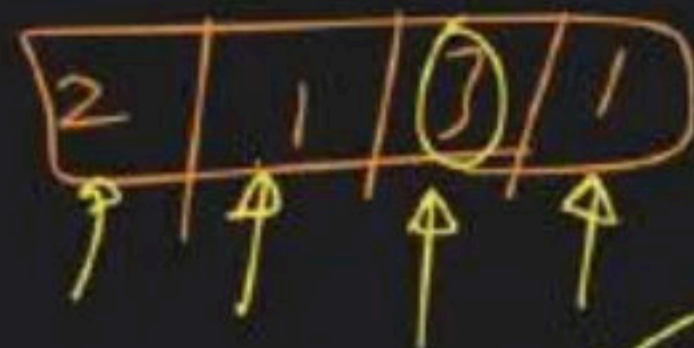
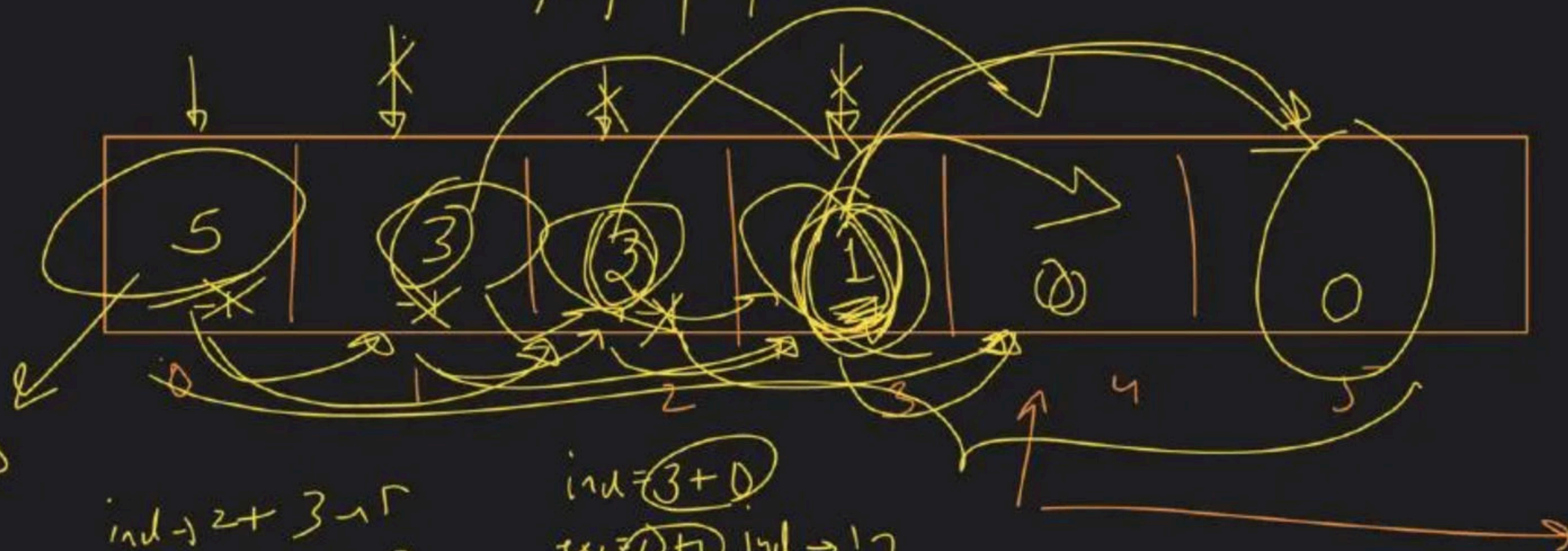
Special class



$\rightarrow$ include $\rightarrow 2 +$ $\text{Rec} \rightarrow \text{Rec}$ $\rightarrow a_4$
 $\text{exclude} \rightarrow 0 +$ $\text{Rec} \rightarrow \text{Rec}$ $\rightarrow a_3$

$\left. \begin{array}{l} \text{include} \rightarrow 2 + \text{Rec} \rightarrow \text{Rec} \rightarrow a_4 \\ \text{exclude} \rightarrow 0 + \text{Rec} \rightarrow \text{Rec} \rightarrow a_3 \end{array} \right\} \rightarrow \text{max} \rightarrow \text{final Ans}$



$$n = 4 \rightarrow$$

$$(n+2) \rightarrow dp$$
 $a \vee b$
$$\text{ind} \rightarrow 2 + 3 + 15$$

$e \times 1 \rightarrow 0 + 7 \rightarrow 7$

$$ind = 1 + 1 - 2$$
$$x \cdot d = 0 + 3 \rightarrow 3$$
$$\text{ind} = 3 + 0$$

ex: $(0+1)$ ind $\rightarrow 1$
ex: 0
ex: 10

$$x \rightarrow 0$$


TD

$f(9)$

do
answer to
=

$f(8)$

$f(7)$

OK hi. Baav

$f(7)$

$f(6)$

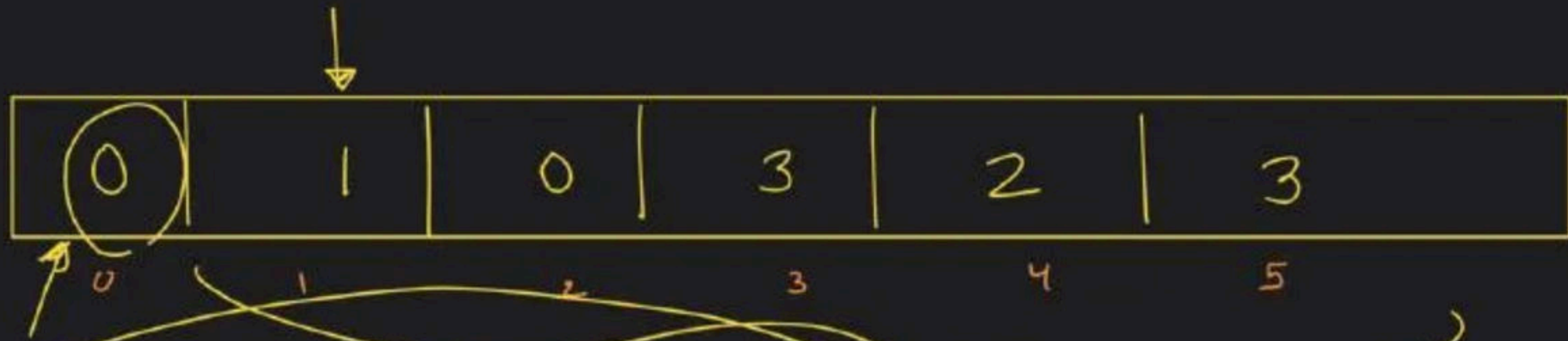
|

|
|
|

BU



7



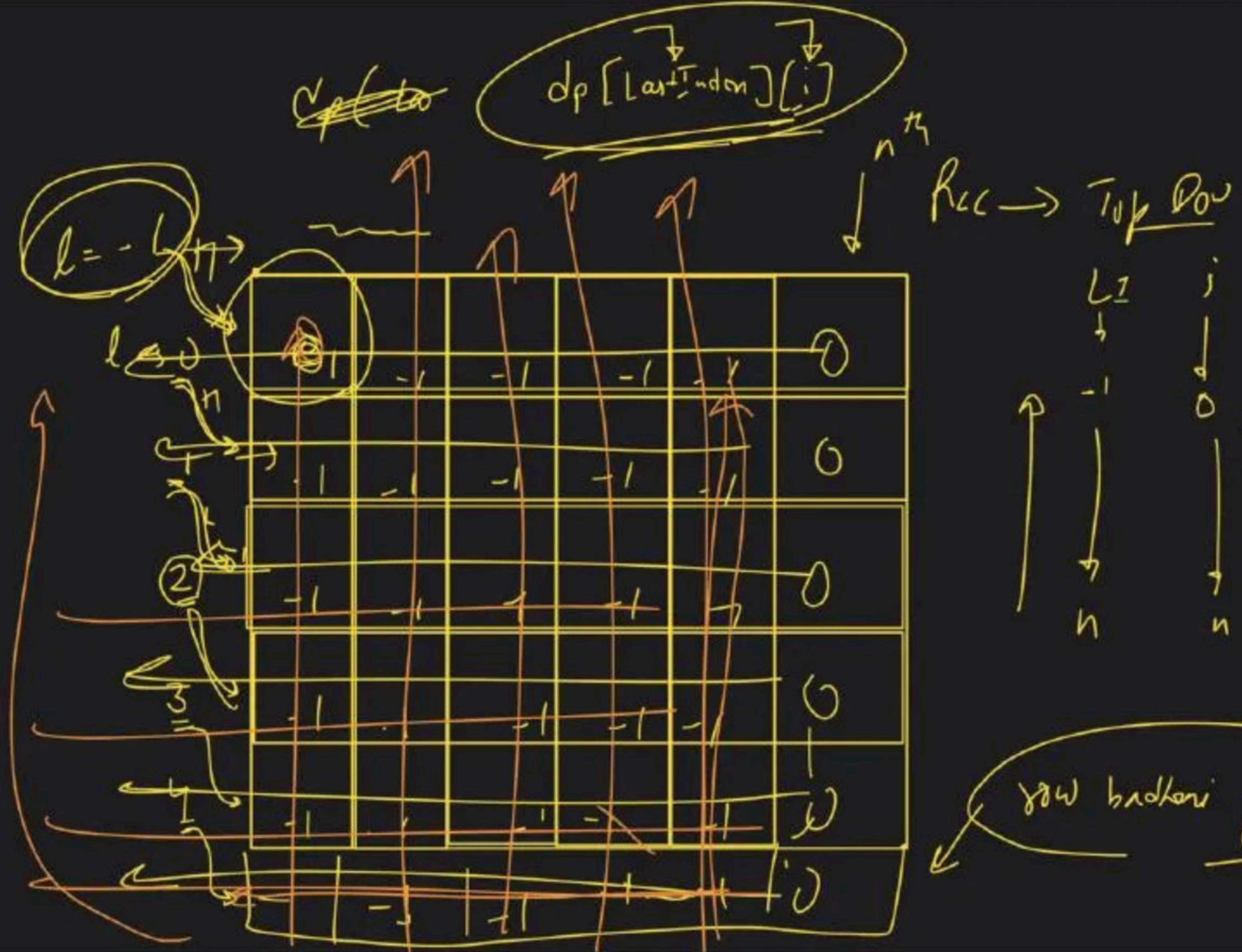
LIS

inc {0} + Rec

calc 0 + Rec

0 → last

NT-MIN



$dp[amt] \rightarrow \underline{dp[amt - coin]}$

1 2 3 4 5
- - - - -
1 2 3 4 5



amt

DP Class - 3

Special class

→ Knapsack Problem → 0/1 → DP
 → Fractional → greedy

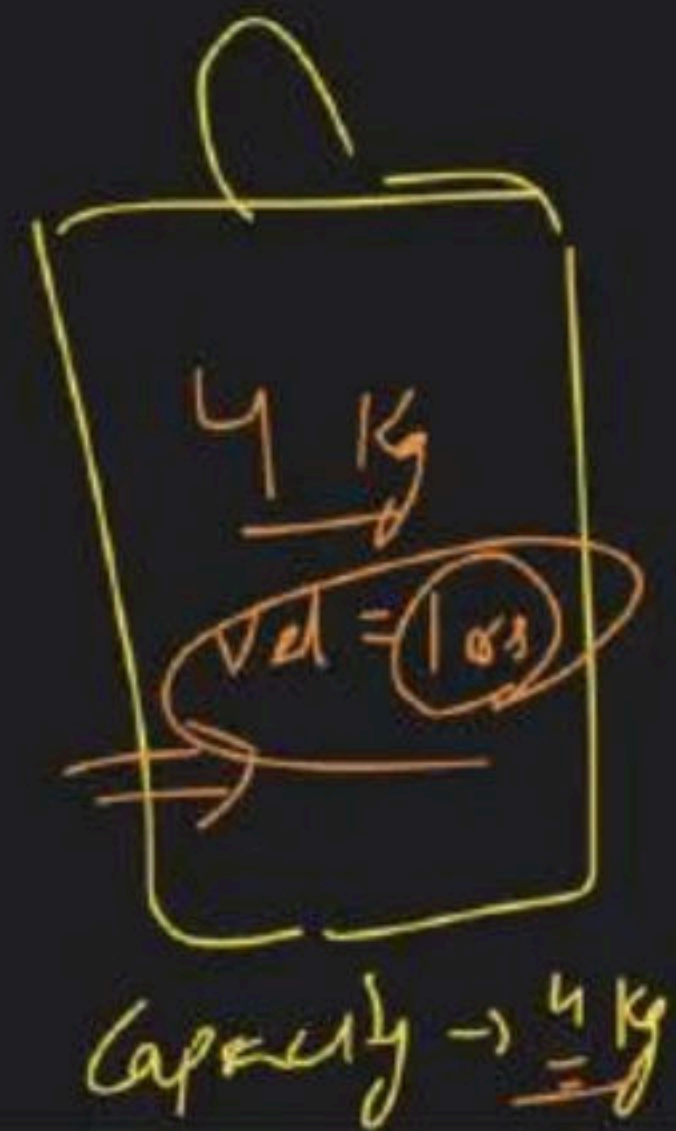
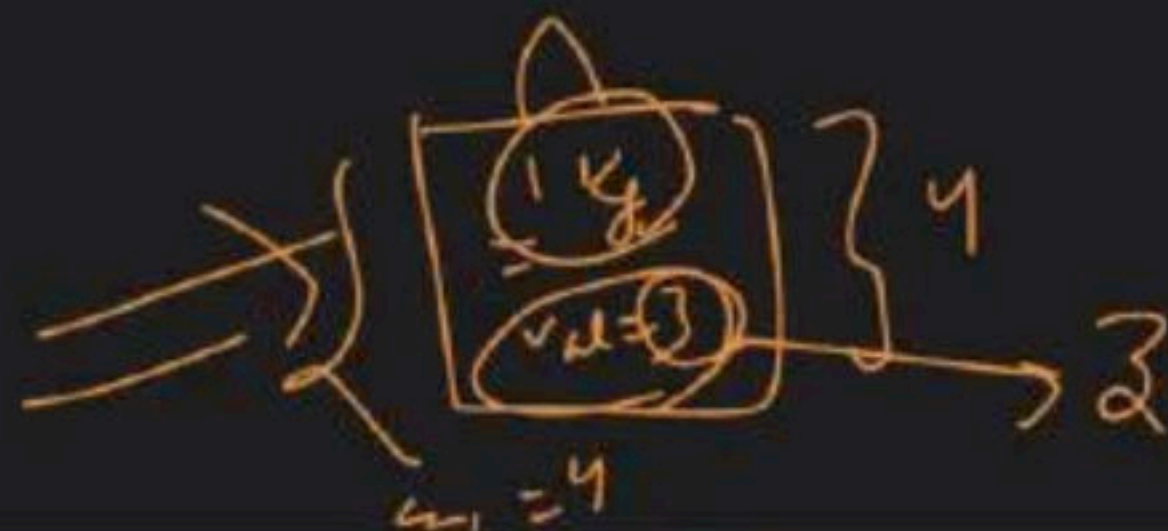


items →

	item 1	item 2	item 3
wt →	{ 4 kg }	{ 5 kg }	{ 1 kg }
val[] →	{ 1 }	{ 2 }	{ 3 }

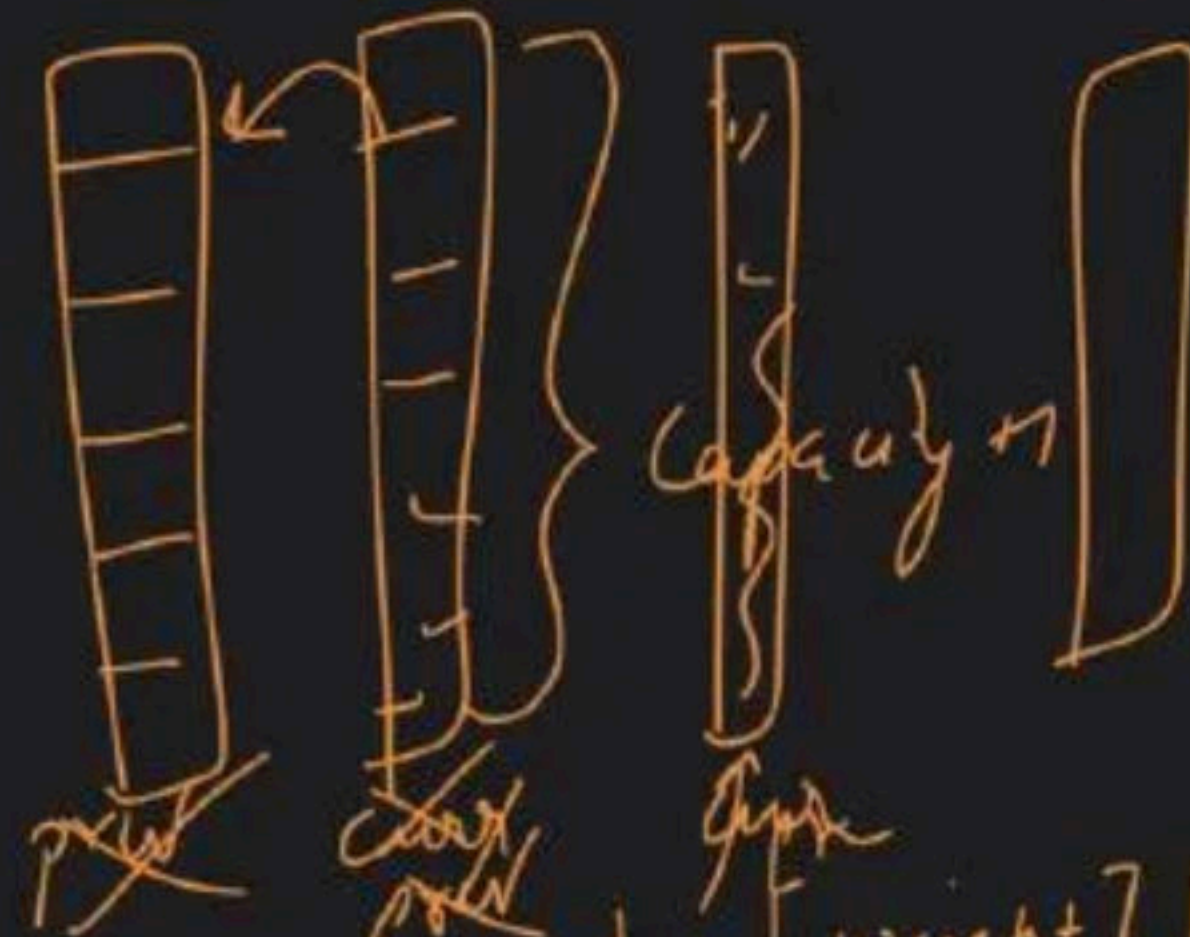
subsequence

inc/excl



S.O

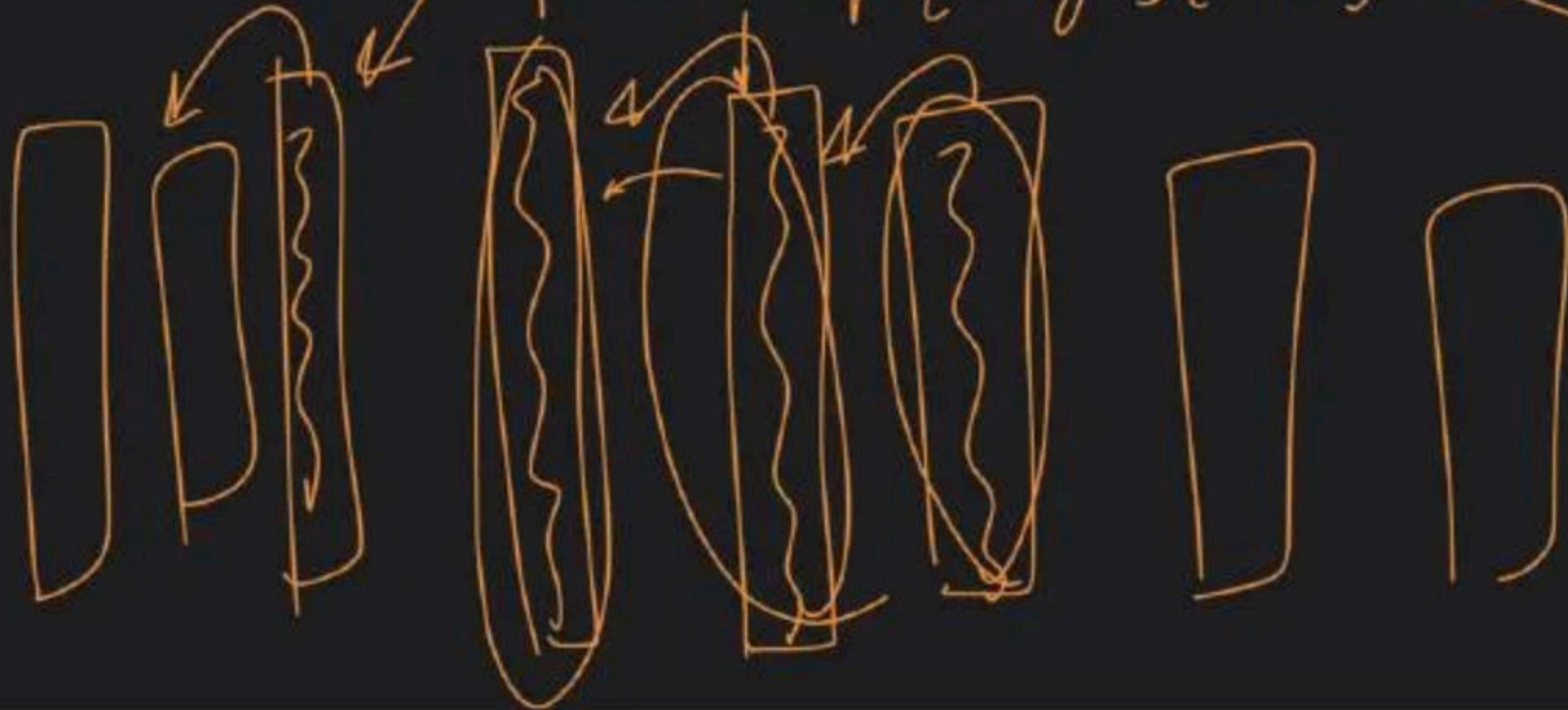
dependency

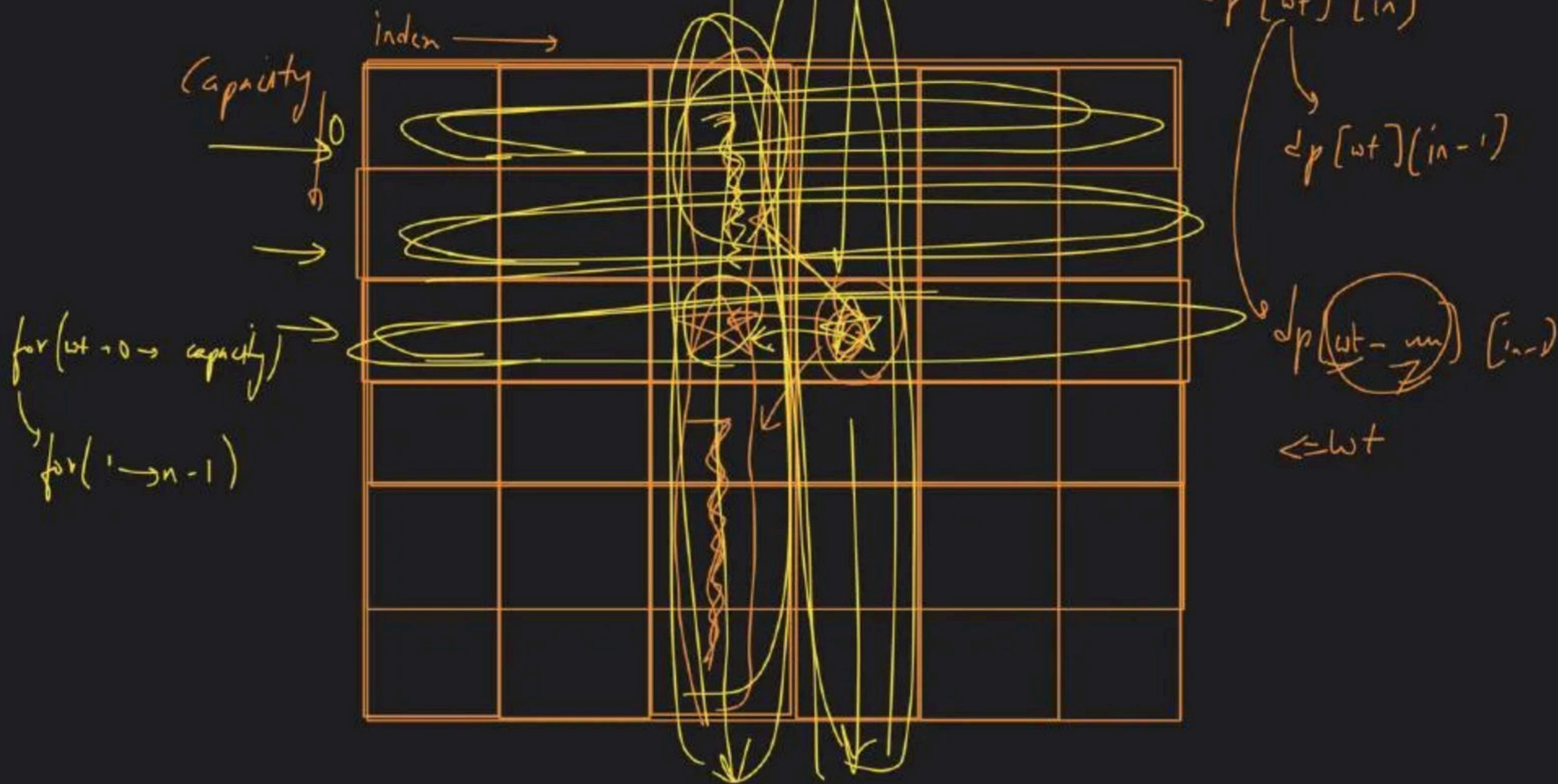


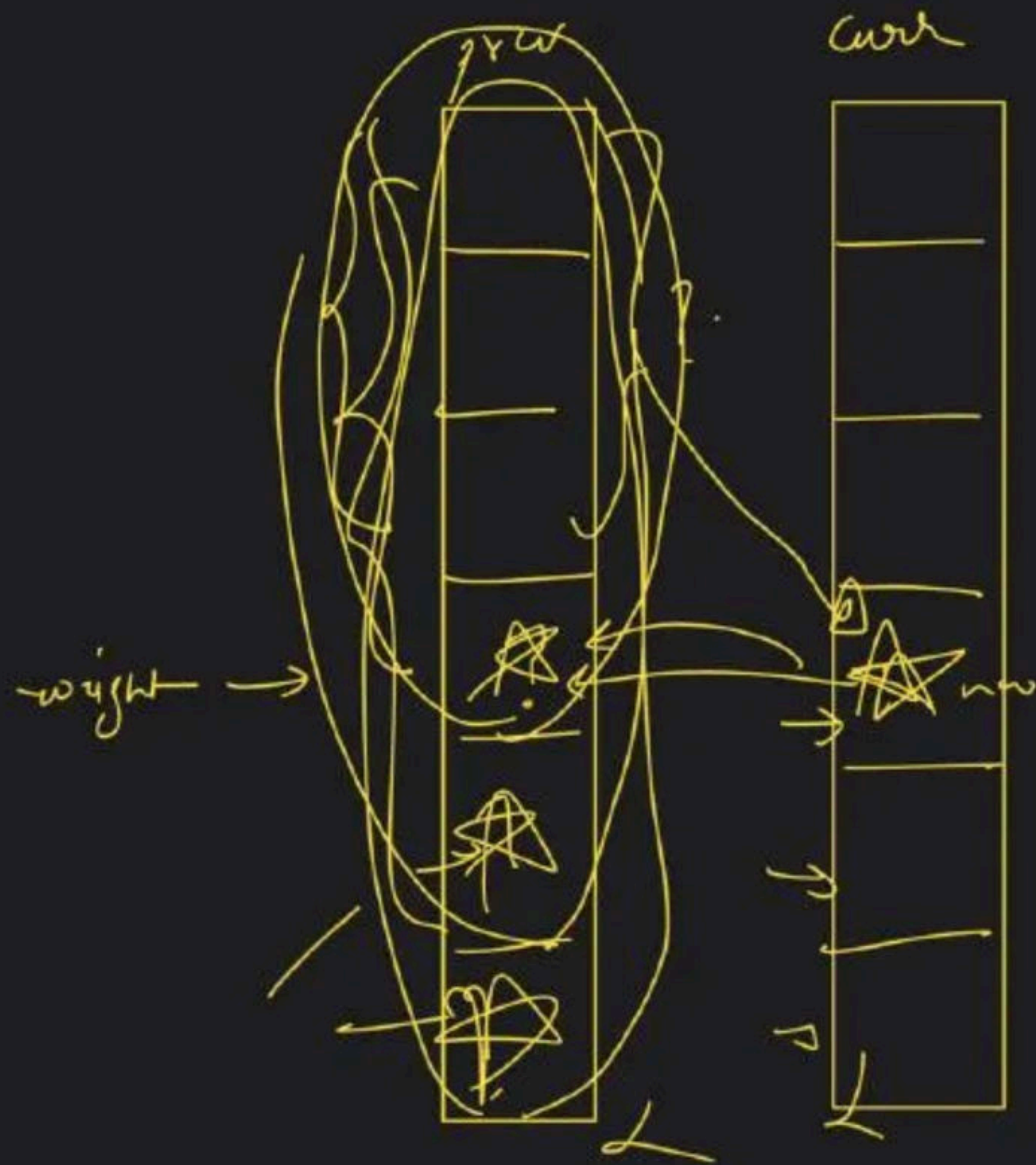
$dp[weight][index-1]$

$dp[weight][index]$

$dp[weight - wt[index]][index-1]$



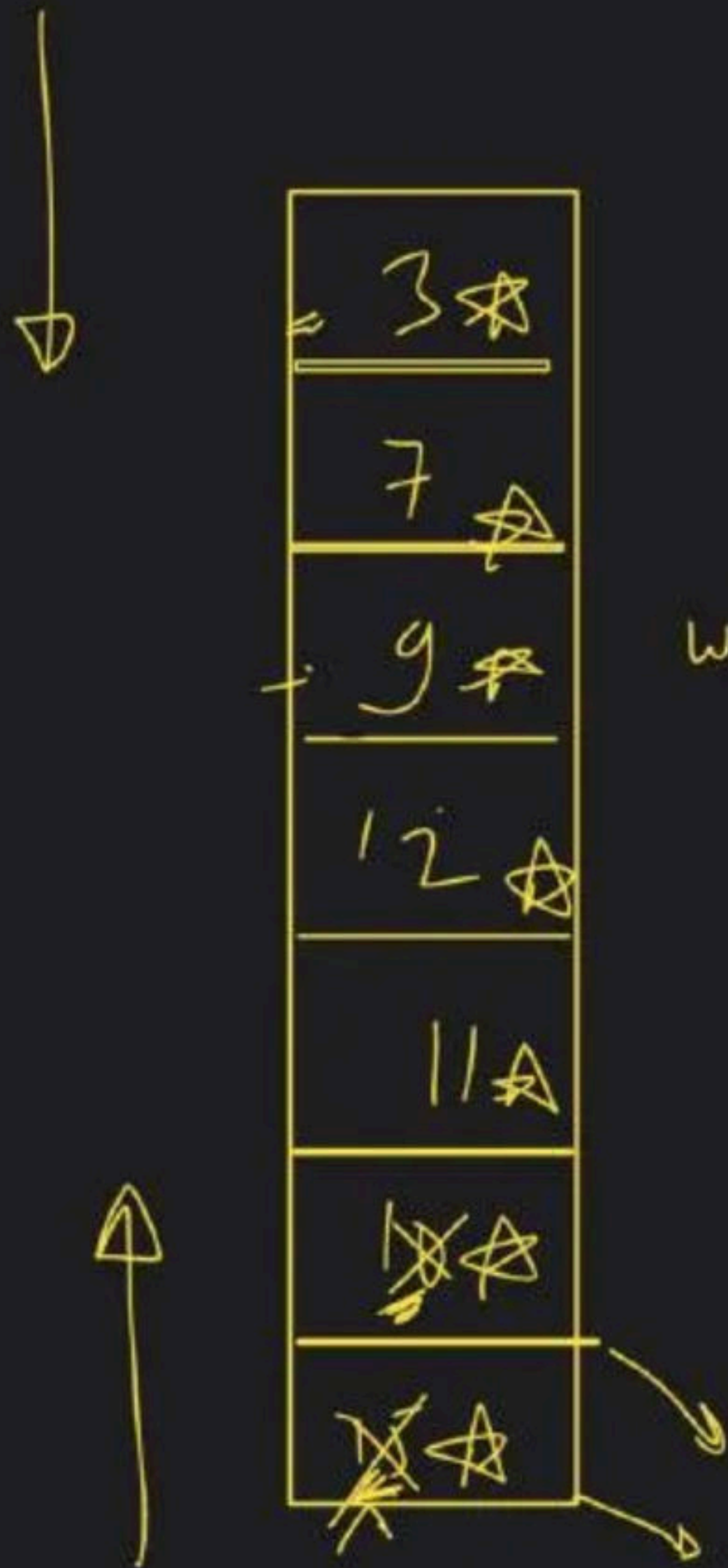




curr [weight]

→ prev [weight]

→ prev [w, w - wt(i)]



~~for~~

for (index = 1 $\rightarrow$ (n-1))

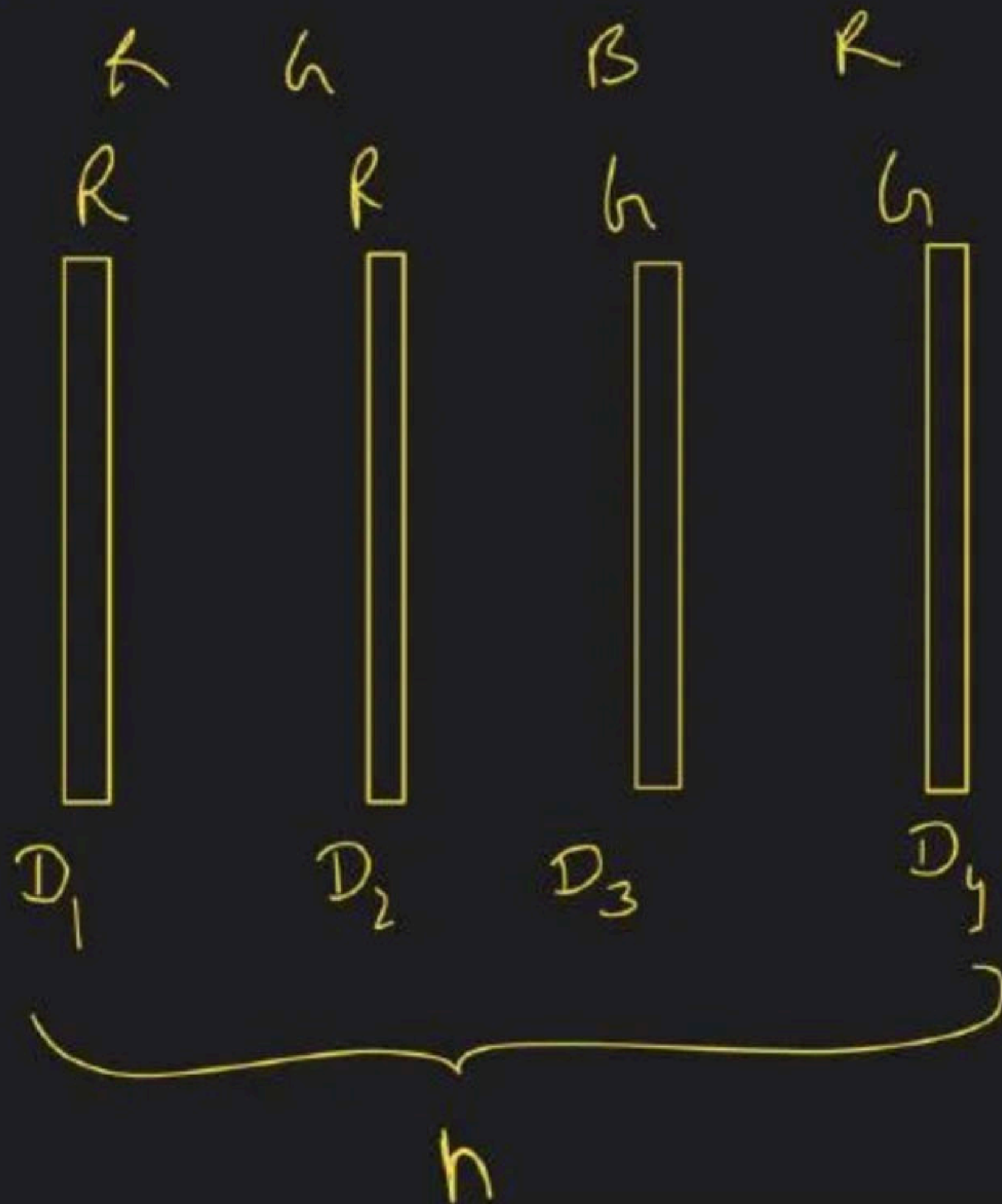
for (weight

0 $\rightarrow$ w)

arr[wight]

-> Painting Fence problem

total no of
ways to paint
the fence



colors $\rightarrow$ (3)
 R, G, B

$R \rightarrow G, B$
 $G \rightarrow R, B$
 $B \rightarrow R, G$

$R \rightarrow R, G, B$
 $G \rightarrow R, G, B$
 $B \rightarrow R, G, B$

$K=3 \rightarrow RLB$

$n=1 \rightarrow 3$

$n=2 \rightarrow 9$

$n=1$

$n=2$

$n=3$

$n=4$

same

R
 RLB

RR
 RLB
 BB

R
 RLB
 BB
 RLB
 BB
 RLB
 BB
 RLB
 BB

RLB
 RLB
 RLB
 RLB
 RLB
 RLB
 RLB
 RLB
 RLB

$f(n-2) * (K-1)$

RLB
 RLB
 BB

$f(n-2) * (K-1)$

RLB
 RLB

RLB
 RLB
 BB

K

diff

RLB
 RLB
 BB
 RLB
 RLB
 RLB
 RLB
 RLB
 RLB

RLB
 RLB
 BB
 RLB
 RLB
 RLB
 RLB
 RLB
 RLB

$f(n-1) * (K-1)$

RLB
 RLB
 BB

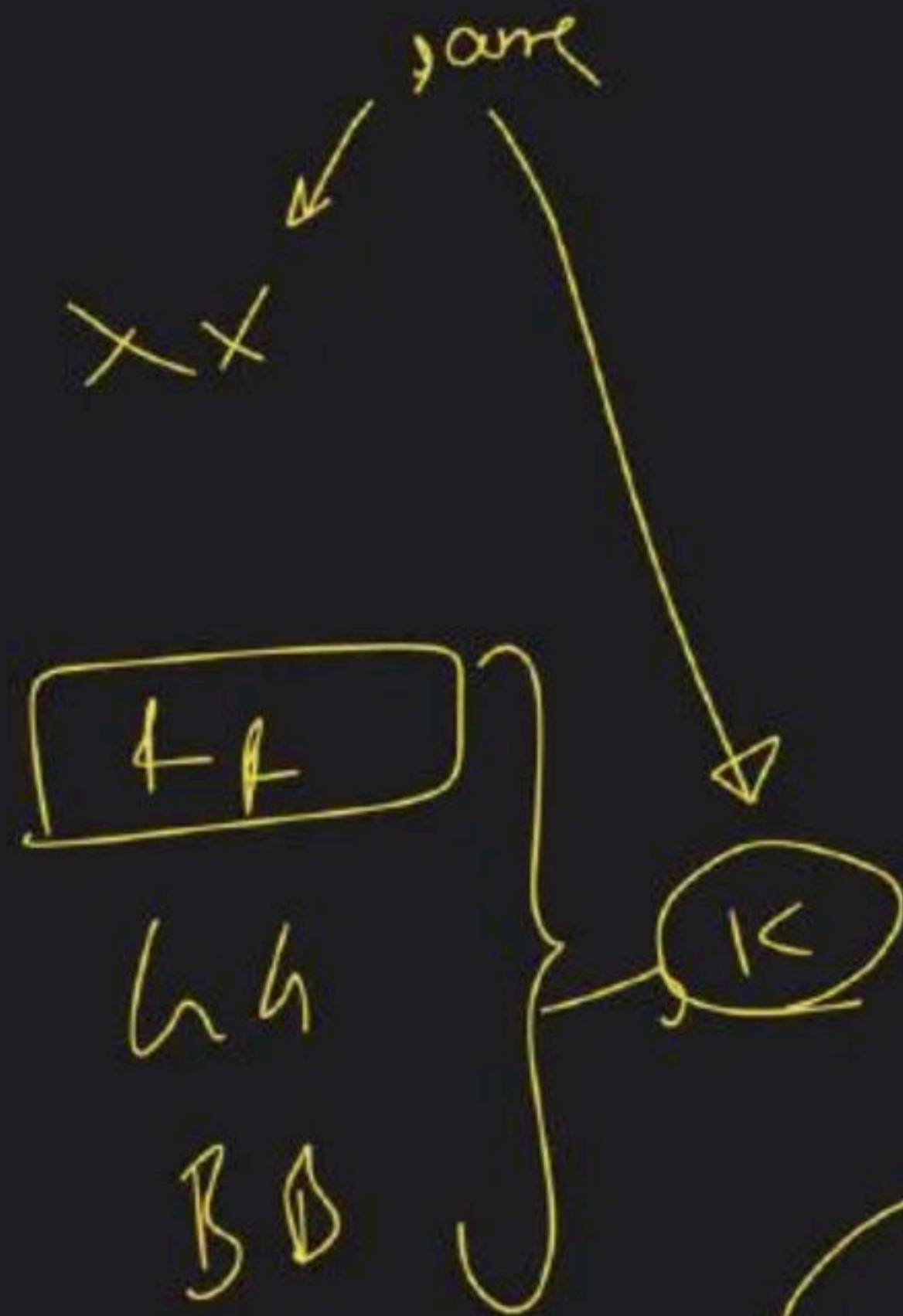
RLB
 RLB
 BB

$f(n-1) * (K-1)$

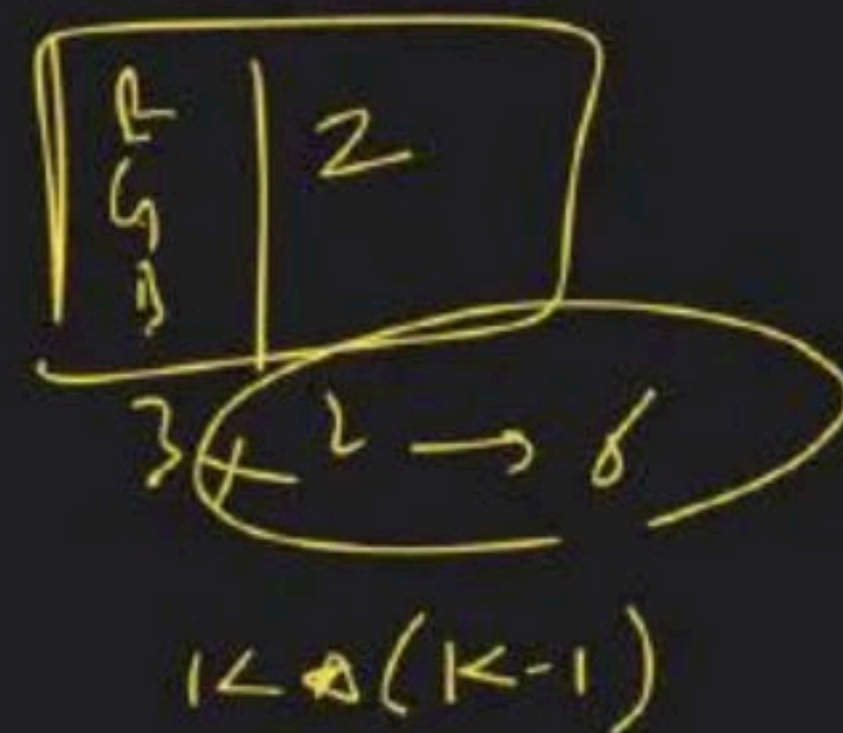
$f(n-2) * (K-1) + f(n-1) * (K-1)$

$(K-1) * (f(n-1) + f(n-2))$

$ans = 3$



X X

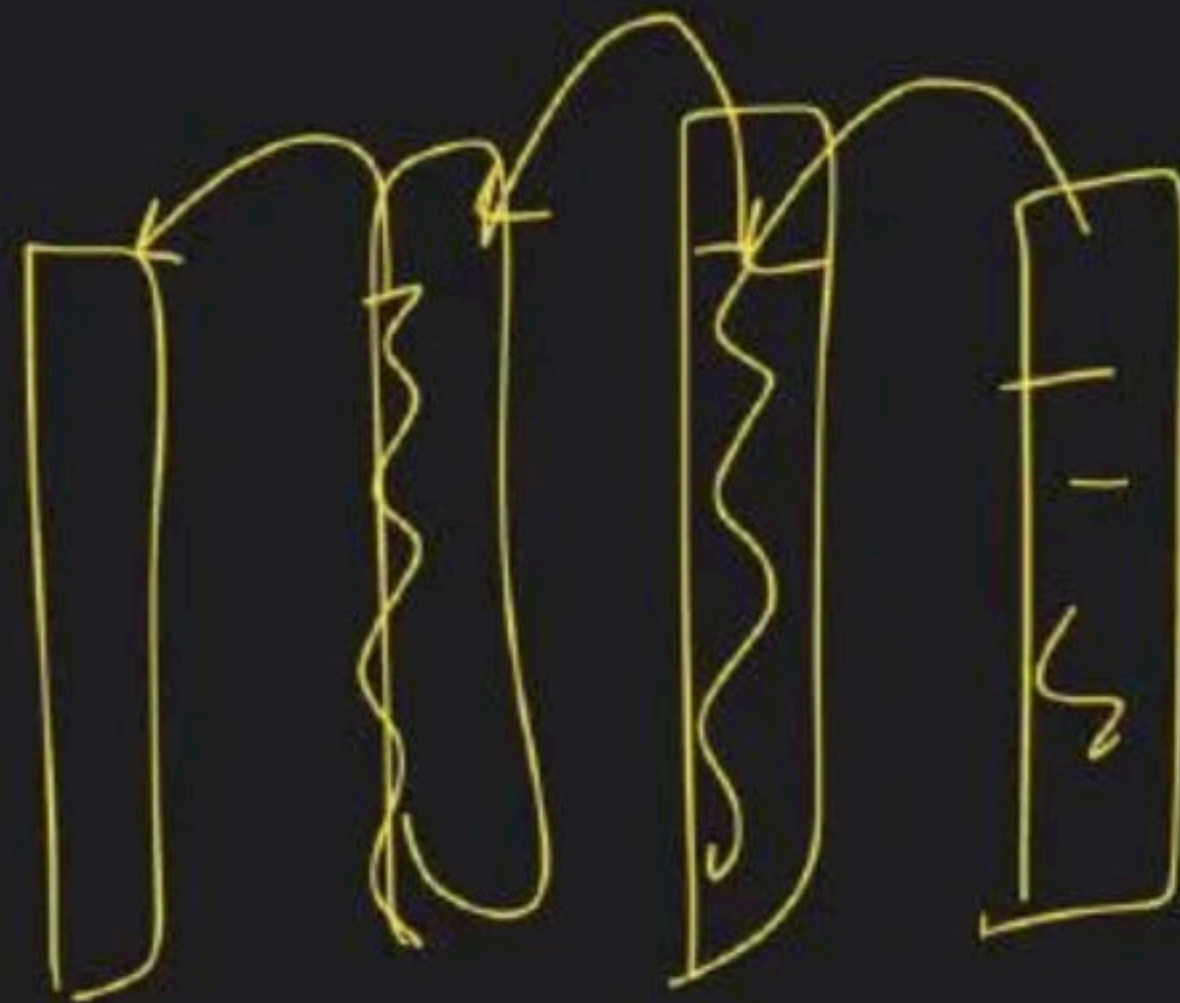
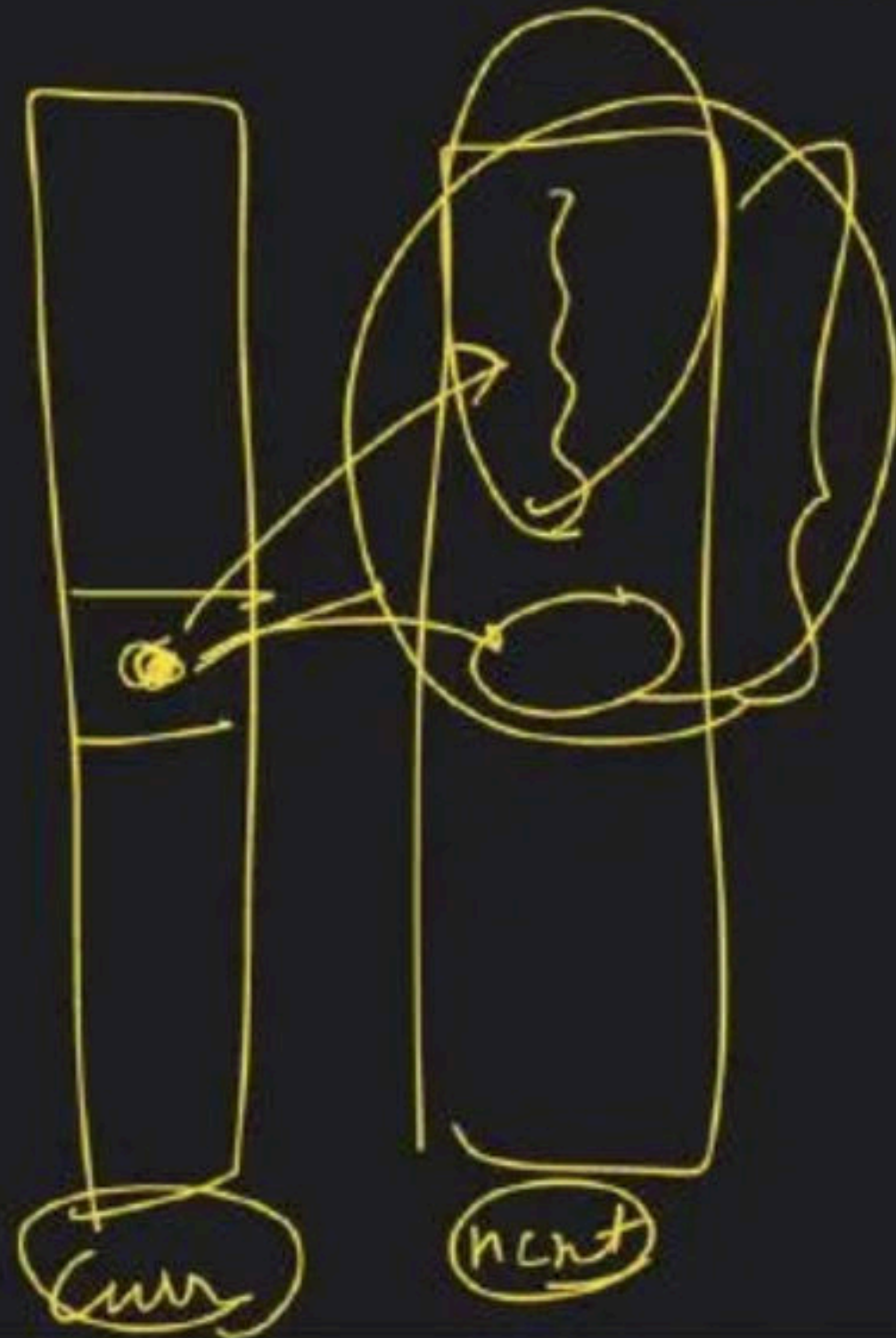


$$K + (K) + (K-1)$$

$dp[t][i]$

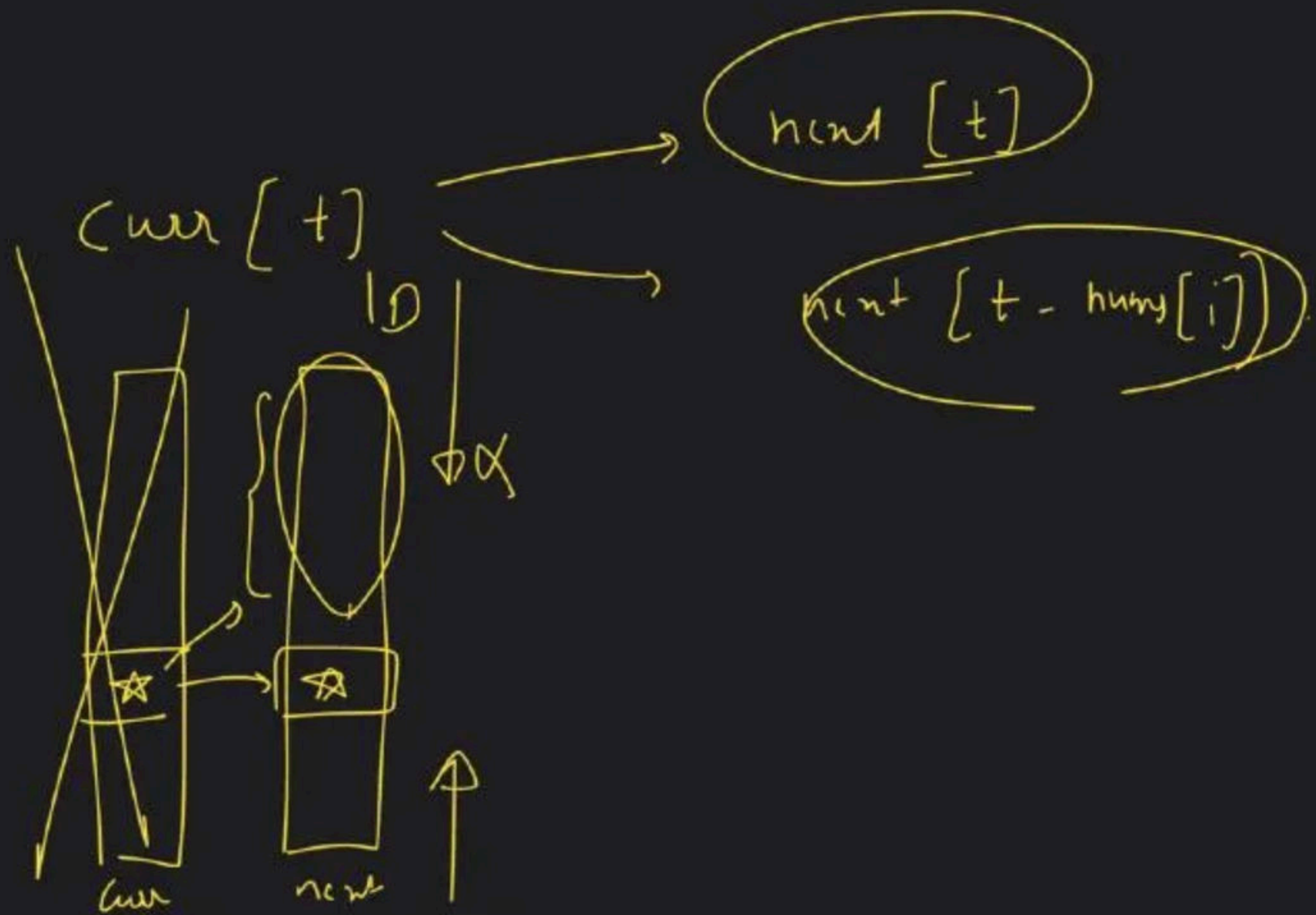
$dp[t][i+1]$

$dp[t - \text{nums}[i]][i+1]$



Partition Equal Subset Sum

Week 1 DP KD
@ 9 PM
DP
W2 DC



→ $n \rightarrow$ no. of dice

$k \rightarrow$ faces

target $\rightarrow$ sum
~~prob~~

total possible
ways

$$n=2$$

$$k=6$$

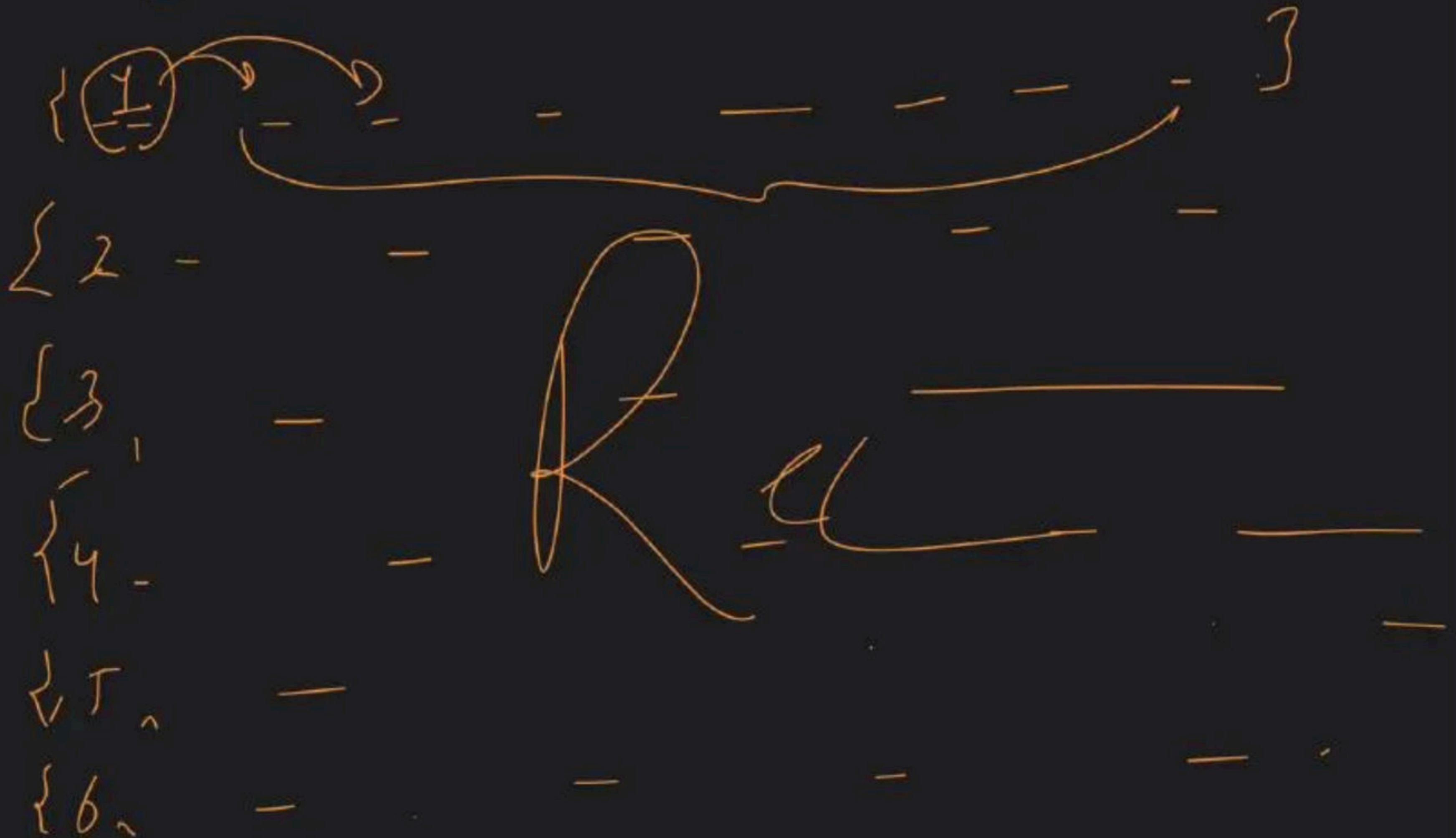
$$t=9$$



4

$\{1, 1\}$	$\{2, 1\}$	$\{3, 1\}$	$\{4, 1\}$	$\{5, 1\}$	$\{6, 1\}$
$\{1, 2\}$	$\{2, 2\}$	$\{3, 2\}$	$\{4, 2\}$	$\{5, 2\}$	$\{6, 2\}$
$\{1, 3\}$	$\{2, 3\}$	$\{3, 3\}$	$\{4, 3\}$	$\{5, 3\}$	$\{6, 3\}$
$\{1, 4\}$	$\{2, 4\}$	$\{3, 4\}$	$\{4, 4\}$	$\{5, 4\}$	$\{6, 4\}$
$\{1, 5\}$	$\{2, 5\}$	$\{3, 5\}$	$\{4, 5\}$	$\{5, 5\}$	$\{6, 5\}$
$\{1, 6\}$	$\{2, 6\}$	$\{3, 6\}$	$\{4, 6\}$	$\{5, 6\}$	$\{6, 6\}$

10 days



$$dp[N][t] \longrightarrow dp[\underline{N-1}][\underline{t-1}]$$

2×2

